Project SG3

What Affects Tennis Win Percentages?

CMSE 202 - 004 | Jackson Wright, Christy Staniak, Nathan Daou, Sophie Scherkenbach, Cheikh Beye

Research Question

Can we predict the winner of a tennis match based on information about the match?

What are the best predictive factors for winning the match?



Datasets and Libraries Used

ATP professional tennis match data (2014 & 2024)

Variables of Interest (match statistics):

- Aces
- Double faults
- 1st & 2nd serve points won
- Break points saved

Contains data for both winners and losers

Python Libraries

- NumPy and Pandas for data processing
- Matplotlib for plotting
- Scikit-learn for various machine learning models and tools

Data Cleaning and Preparation

We merged the 2014 and 2024 datasets since they shared the same schema

We also dropped NaN values and some columns that we chose not to use

All remaining columns had numeric data types

We filtered the data to only include rows from the four major tournaments (Australian Open, US Open, Wimbledon, Roland Garros)

Feature Engineering

Many features in the dataset had variants for the winner and loser of each match

We compressed these features into difference features (winner feature minus loser feature)

We trained our models on these difference features because they encode information about the match rather than the players themselves

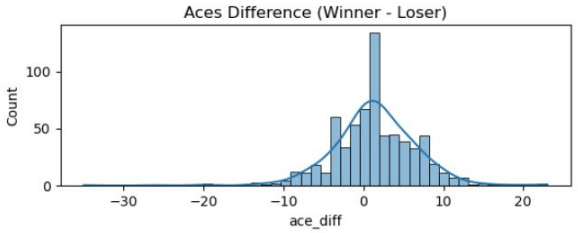
Doing it this way led to much better results

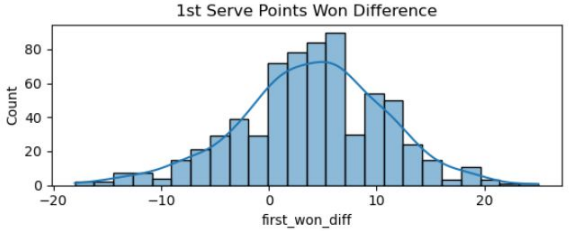


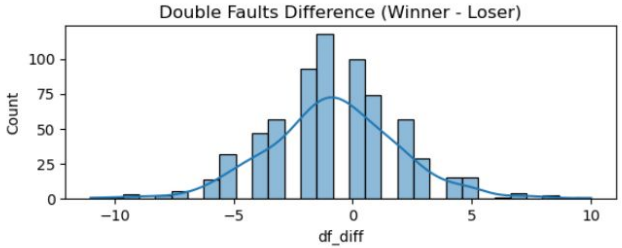
Results of Feature Engineering

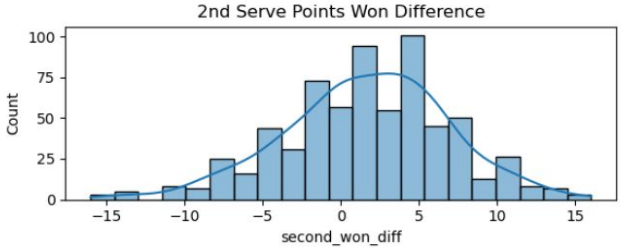
For serve points difference and aces, the winners on average outperformed the losers, as shown by the graphs centering on positive x values

For double faults, the losers on average outperformed the winners, as shown by the center of the graph being centered on a negative x value









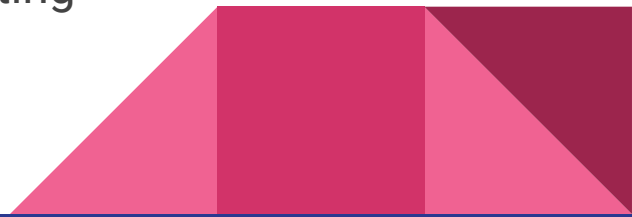
Samples and Labels

To create samples with labels, we labeled all the regular matches with these features as winner, and we created negative variants (loser - winner) and labeled them as loser

The features for each were player stats relative to their opponent’s stats

Basically, our models learned to predict whether or not player A won the match by looking at player A’s stats - player B’s stats

We used 80% of the samples for training and 20% for testing

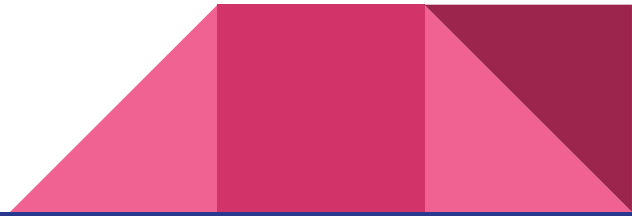


Models

Models We Tried:

- Logistic Regression
- SVM
- Random Forest
- KNN

We encoded our labels appropriately for each model, and various combinations of appropriate hyperparameters were tested for each model.

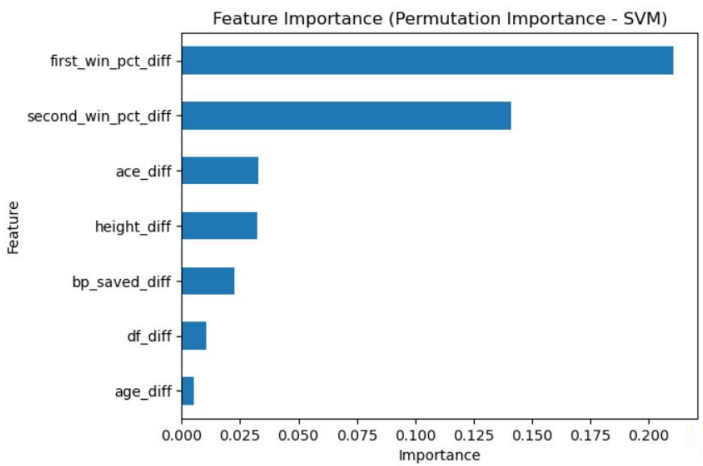


Results of Models

The performance of the best model in each category was evaluated using accuracy on the test set as the evaluation metric.

- **Logistic Regression**
  - 91.9% test accuracy
- Random Forest
  - 86.7% test accuracy
- **SVM**
  - 88.6% test accuracy
- KNN
  - 62.7% test accuracy

Results of Models cont.



Model accuracy: 93.57%

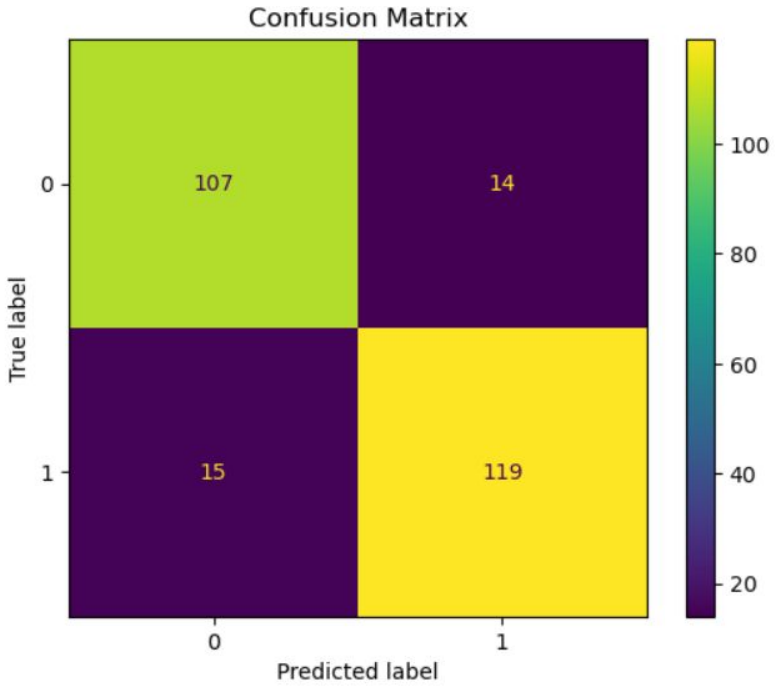
Model coefficients:

df_diff	-0.059105
age_diff	-0.002282
height_diff	-0.001110
ace_diff	0.039312
bp_saved_diff	0.091784
second_win_pct_diff	12.130153
first_win_pct_diff	17.093815

Permutation importance analysis revealed that second serve performance and first serve effectiveness were the most influential features in predicting match outcomes

Losgistic regression model confirms this.

Results of Models cont.



Confusion matrix shows that there is a high volume of true negatives and positives, and low values of false positives and negatives

This shows that our model is very accurate

Discussion and Conclusion

Logistic Regression performed best (~92% accuracy)
Match outcomes are highly predictable using performance stats
Key factors influencing wins:

- More aces
- Fewer double faults
- Strong 1st & 2nd serve performance
- Effective break point play

Serve performance is the most important factor in winning
Player characteristics (age, height) have little impact
Overall: **In-match performance matters more than physical traits**